

PACKAGE LEAFLET**MONIMAX®****80 mg/g + 80 mg/g premix for medicated feeding stuff****STATEMENT OF ACTIVE SUBSTANCE:**

Each kilogram contains 80 g monensin (as monensin sodium) and 80 g of nicarbazin.

PRODUCT DESCRIPTION:

MONIMAX® premix for medicated feeding stuff represents green-brown granules.

PHARMACODYNAMICS:

Monensin sodium acts as an ionophore, i.e. a chemical substance that complexes monovalent cations with the lipophilic structure that facilitates the transport of the bound ion through biological membranes. Coccidial sporozoites exposed to monensin sodium in the intestinal lumen exhibit considerable swelling, large vacuoles, pitting and holes in the surface suggesting extreme osmotic damage which is potentially lethal. Development of a sporozoite that successfully invades a host cell is inhibited as the ionophore continues its destructive process. Monensin sodium selectively destroys intracellular sporozoites while remaining relatively non-injurious to the host cell (Chapman, 1993). It has been shown that monensin sodium has an effect on second generation merozoites but not upon developing gametocytes.

Monensin sodium is primarily coccidiocidal in action and active against *Eimeria acervulina*, *E. brunetti*, *E. maxima*, *E. mivati*, *E. necatrix* and *E. tenella* of the chicken and *E. melagrimitis*, *E. pallida*, *E. dispersa* and *E. adenoides* of the turkey. Nicarbazin, a chemically synthesised product, is an equimolecular complex of 4,4'-dinitrocarbanilide (DNC) and 2-hydroxy-4,6-dimethylpyrimidine (HDP). DNC, not HDP, has anticoccidial activity when used alone against *Eimeria*. The potency of DNC is increased ten fold when this molecule is complexed with HDP as nicarbazin, but no increase in anticoccidial activity is observed with a simple mixture of the two compounds. Dissolution of the DNC-HDP complex in the gut results in the formation of much smaller DNC crystal size than if the water insoluble DNC is administered alone that would explain an increased efficacy. Nicarbazin is at least partially broken down in the gut to DNC and HDP which may be absorbed independently.

PHARMACOKINETICS:

Monensin sodium is absorbed at a limited extent and excreted rapidly, it is extensively metabolized and gives rise to demethylated, oxidised and decarboxylated metabolites.

Nicarbazin, when ingested, is rapidly split in its two components dinitrocarbanilide (DNC) and 2-hydroxy-4,6-dimethylpyrimidine (HDP) which behave independently. Liver is the target tissue. DNC residues decline rapidly from tissues following nicarbazin withdrawal. DNC appears as the marker residue. HDP-related residues are much lower than those derived from DNC. For both compounds of MONIMAX®, the metabolic pathways in the chicken are similar to those in the turkey and rat.

INDICATIONS FOR USE (SPECIFYING THE TARGET SPECIES):

Prevention of coccidiosis caused by *Eimeria* species in chickens for fattening, chickens reared for laying, and turkeys.

AMOUNTS TO BE ADMINISTERED:

To obtain a uniform distribution of the premix in finished feeds, incorporate MONIMAX® into a secondary premix prior to mixing it into the finished feed. Clean all mixing equipment thoroughly before preparing other feeds.

Species	Minimum content	Maximum content	Quantity of MONIMAX® incorporated into the feed (g/ton)	Maximum age
	mg of active substance / kg of complete feeding stuff with a moisture content of 12%			
Chicken for fattening	40 mg	50 mg	500 - 625	-
Chickens reared for laying	monensin 40 mg nicarbazin	monensin 50 mg nicarbazin		16 weeks
Turkeys				16 weeks

ADMINISTRATION ROUTE:

For oral administration through the feed.

CONTRAINDICATIONS:

Dangerous for equines. This coccidiostat contains monensin so simultaneous administration of tiamulin should be avoided at least seven days before or after receiving feed containing MONIMAX®. Monitor for possible adverse effects when used concurrently with other medicinal substances. MONIMAX® cannot be used in chickens reared for laying and turkeys for fattening over 16 weeks of age.

ADVERSE REACTIONS (FREQUENCY AND SERIOUSNESS):

The coccidiostat MONIMAX® (monensin sodium and nicarbazin) is considered safe for chickens for fattening and chickens reared for laying at the highest use level of 50 mg monensin and 50 mg nicarbazin per kg of complete feed.

WARNINGS AND PRECAUTIONS:**Special precautions for use in animals:**

Do not administer to chickens reared for laying over 16 weeks of age.

Do not administer to turkeys over 16 weeks of age.

Do not feed undiluted to animals.

Special precautions for the person administering the veterinary medicinal product to animals:

Harmful if swallowed. Harmful if inhaled. Causes serious eye irritation.

Causes skin irritation. Avoid breathing dust. Use only outdoors or in a well-ventilated area. Wash hands thoroughly after handling. Do not eat, drink or smoke when using this product. Operators, handling the product, to wear face mask, gloves and protective clothing and to avoid inhalation of and contact with the product. Wash hands and exposed skin after handling the product. If MONIMAX® enters the eyes, rinse out with water immediately.

USE DURING PREGNANCY, LACTATION OR LAY:

MONIMAX® cannot be used in chickens reared for laying and turkeys for fattening over 16 weeks of age.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION:

MONIMAX® shall not be mixed with other coccidiostats

OVERDOSE (SYMPTOMS, EMERGENCY PROCEDURES, ANTIDOTES), IF NECESSARY:

Chickens: MONIMAX® at the highest proposed dose (50 + 50 mg monensin + nicarbazin/kg complete feed) is considered safe for chickens for fattening with a margin of safety of about 1.5.

Turkeys: MONIMAX® is considered safe for turkeys for fattening at the highest use level of 50 mg monensin and 50 mg nicarbazin/kg complete feed. The margin of safety is about 1.5.

WITHDRAWAL PERIODS:

Chickens for fattening: Meat: 0 day.

Chickens reared for laying: Meat: 0 day.

Turkeys: Meat: 0 day.

SPECIAL PRECAUTIONS FOR THE DISPOSAL OF UNUSED PRODUCT OR WASTE MATERIALS, IF ANY:

Prevent spilled material from flowing onto adjacent land or into streams, ponds or lakes.

Methods and material for containment and cleaning up:

Carefully sweep the powder up and remove. Place material in a dry container and cover. Remove from the area. Flush spill area with water.

STORAGE CONDITIONS:

Store below 30°C. Store in a dry place.

Store in the original packaging, well closed, in dry and well ventilated facilities, protected from direct sunlight. Expiry date is 2 years from the date of manufacture. No influence due to pelleting is expected. The product will remain stable in the finished feedingstuff for 3 months and in a premix for a period of 6 months.

Keep out of the sight and reach of children.

PACKAGING AVAILABLE:

Paper bag with internal polyethylene layer of 20 kg.

MANUFACTURER:

BIOVET Joint Stock Company

39 Petar Rakov Str.

4550 Peshtera, Bulgaria

MARKETING AUTHORIZATION HOLDER:

Pahang Pharmacy Sdn. Bhd. (197101000448)

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DATE OF REVISION:

December, 2024